HDL Design - NUIST

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

HDL Design - NUIST (A35622)

Short Title:	HDL Design (NUIST)		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Discipline Specific Technologies		
Author	CLAIRE.FITZPATRICK		
Credits	5	**Level:**	Postgraduate

Description of Module / Aims

This module aims to provide the student with a working knowledge of digital system design, capture and verification using the VHDL/Verilog Hardware Description Languages and implementation of design on CPLD or FPGA.

Programmes

`HDLD-0001`	MEng in Electronic Engineering (`WD_EELEN_R`)	1 / 2 / M
	MEng in Electronic Information Systems (`WD_EELEN_RI`)	2 / 4 / M
`HDLD-0001`	PG Dip in Engineering in Electronic Engineering (`WD_EELEN_G`)	1 / 2 / M

Indicative Content

- Programmable Logic Architecture
- VHDL/Verilog Language Concepts
- Signals, Data Types, Operate
- Behavioural and register transfer level modelling
- Structural modelling
- Concurrent execution: Procedures and Functions, sequential statements
- Design verification (pre and post synthesis)
- VHDL/Verilog Synthesis and implementation
- Design, Capture, synthesis verification and implementation of special purpose digital system

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Comprehend behavioural, register transfer level and structural HDL based modelling, design capture and simulation and apply to the development of digital systems.
2. Synthesise behavioural, register transfer level and structural HDL modules.
3. Write efficient testbench code to analyse the integrity of the design at all stages.
4. Apply VHDL/Verilog language principles to the design of complex digital systems.

Learning and Teaching Methods

- Lectures
- Assignments Design, capture and design verification problems will be assigned throughout the module to reinforce the concepts demonstrated in class.
- Mini Project

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Assignment	50%	1,2,3
Project	50%	1,2,3,4

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

40% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	6	
Independent Learning	105	

Supplementary Material

- Ashenden, P.J. _...The Designers Guide to VHDL_. 3rd Ed.. USA: Morgan Kaufmann, 2008.
- Pedroni, V.A. _...Circuit Design & Simulation with VHDL_. 2nd Ed.. USA: MIT Press, 2010.

Requested Resources

- Lecture Room: Loose Seated